

Multiple Choice (10 marks)

1. 10 calories of heat are supplied to 1 g of water. How much does the mass of the water increase?
 - (a) $5.5 \times 10^{-13}g$
 - (a) $2.5 \times 10^{-13} g$
 - (b) $2.0 \times 10^{-12}g$
 - (b) $4.7 \times 10^{-13} g$
 - (c) $8.6 \times 10^{-13}g$
2. Compute the mean molecular speed v in the light gas hydrogen (H_2) in m/s
 - (a) 1750.0
 - (b) 1500.0
 - (c) 2200.0
 - (d) 2100.0
 - (e) 1400.0
3. Consider two mercury barometers, one having a cross-sectional area of 1 cm² and the other 2 cm². The height of mercury in the narrower tube is
 - (a) two-thirds
 - (b) twice
 - (c) the same
 - (d) half
 - (e) four times higher
4. Which one of these constellations is not located along the Milky Way in the sky?
 - (a) Virgo
 - (b) Perseus
 - (c) Ursa Major
 - (d) Sagittarius
 - (e) Aquila
5. An object, starting from rest, is given an acceleration of 16 feet per *second*² for 3 seconds. What is its speed at the end of 3 seconds?

- (a) 48 ft per sec
- (b) 64 ft per sec
- (c) 30 ft per sec
- (d) 72 ft per sec
- (e) 16 ft per sec

True / False (10 marks)

Answer True or False

- 6. True or False: The angular magnification of the refracting telescope can be calculated as 20, based on the separation of the lenses.
- 7. True or False: The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately $1.0 \text{ GeV}/c^2$.
- 8. True or False: The efficiency of a device that converts 100 J of input energy into 40 J of useful work is 25%.
- 9. True or False: Doubling the thickness of a neutral density filter will always halve its transmittance regardless of initial transmission values.
- 10. True or False: If the volume of an object doubles while its mass remains constant, its density will halve.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the shape of the illuminated area on a screen positioned 1 cm and 2 m away from a tiny plane mirror measuring $2 \text{ mm} \times 2 \text{ mm}$, explaining the underlying physical principles.
- 12. Considering a converging lens with a 5.0 cm focal length used as a simple magnifier, determine the object distance required to produce a virtual image 25 cm from the eye. Additionally, calculate the magnification and discuss the implications of these values in the context of optical systems.

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